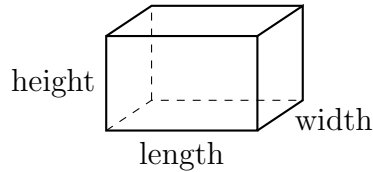


Measurement Conversion, Volume, and Angle Measure

Unit conversion, prism volume, angle measure, and multistep synthesis

Directions: Read each problem for the units it gives you and the unit it asks for. Show the conversion, the formula, or the angle relationship you used, then write the answer with its unit on the answer line. Useful facts: 1 ft = 12 in, 1 m = 100 cm, 1 kg = 1000 g, 1 L = 1000 mL, 1 h = 60 min, and 1 L of water fills 1000 cm³.



How every box on this sheet is labelled. No values shown: use the numbers given in each problem.

- 1.** A jump rope is 4 ft long. How long is the rope in **inches**?

Answer: _____ in

- 2.** A water bottle holds 3 L. How many **millilitres** does it hold?

Answer: _____ mL

- 3.** A gift box is a rectangular prism 8 cm long, 5 cm wide, and 3 cm high. Find its **volume**.

Answer: _____ cm³

4. Ray BD lies between ray BA and ray BC , so angles ABD and DBC together make angle ABC . If $\angle ABD = 35^\circ$ and $\angle DBC = 47^\circ$, find $\angle ABC$.

Answer: _____ degrees

5. A baker weighs out 2 kg 250 g of flour. What is the total mass in **grams**?

Answer: _____ g

6. A storage bin is a rectangular prism with a volume of 240 cm^3 . Its base is 8 cm by 6 cm. How **tall** is the bin?

Answer: _____ cm

7. Angle PQR is a right angle. Ray QS splits it into $\angle PQS$ and $\angle SQR$. If $\angle PQS = 34^\circ$, find $\angle SQR$.

Answer: _____ degrees

8. A film runs for 2 h 25 min. How long is the film in **minutes**?

Answer: _____ min

9. A shipping carton measures 12 cm by 10 cm by 4 cm. Wooden cubes with an edge of 2 cm are packed inside with no gaps. How many cubes fill the carton? (Find the carton's volume and one cube's volume first.)

Answer: _____

10. Rosa measured an angle that is clearly *acute* — both of its rays open less than a square corner — but she wrote down 130° . Her protractor shows two scales. Explain what Rosa most likely did wrong and describe one check she can make before writing an angle measure down. Then give the angle's true measure.

Answer: _____ degrees

11. A fish tank is 40 cm long, 25 cm wide, and 30 cm high. When it is filled to the top, how many **litres** of water does it hold? (1 L = 1000 cm^3)

Answer: _____ L

12. A packing crate is 2 ft long, 18 in wide, and 1 ft high. Find its volume in **cubic feet**. (Convert every edge to feet first.)

Answer: _____ ft^3